

LIBRARY MANAGEMENT SYSTEM

Project Title

Library Management System

Introduction

The Library Management System is a C++ based application developed to manage library operations efficiently. It helps in maintaining book records, searching books, issuing books to users, and returning books using a simple menu-driven interface.

Objective

To develop a system that automates basic library functions such as adding books, searching books, issuing and returning books, and displaying available records.

Concepts Used

- Structures
- Functions
- Arrays
- Loops
- Conditional Statements
- Menu-Driven Programming
- Object-Oriented Programming Basics

Features

- Add new books
- Display all books
- Search books by title or author
- Issue books
- Return books
- Check availability status

Technologies Used

- Programming Language: C++
- IDE: VS Code / Turbo C++
- Compiler: GCC Compiler

Expected Outcome

The system simplifies library management by storing and maintaining book details efficiently. It reduces manual work and provides quick access to library records.

Advantages

- Easy to use
- Fast record management
- Saves time
- Reduces manual errors
- User-friendly interface

Sample Output

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

```
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
```

```
Enter Your Choice: 1
```

```
Enter Book ID: 101
Enter Book Title: C++ Programming
Enter Author Name: Bjarne Stroustrup
```

```
Book Added Successfully!
```

```
Enter Your Choice: 2
```

```
===== BOOK LIST =====
```

```
Book ID: 101
Title: C++ Programming
Author: Bjarne Stroustrup
Status: Available
-----
```

```
Enter Your Choice: 4
```

```
Enter Book ID to Issue: 101
```

```
Book Issued Successfully!
```

```
Enter Your Choice: 5
```

```
Enter Book ID to Return: 101
```

```
Book Returned Successfully!
```

```
Enter Your Choice: 6
```

Thank You!

Conclusion

The Library Management System is a simple and effective mini project that demonstrates the practical implementation of C++ programming concepts for managing library operations efficiently.